

Total No. of Questions : 8]

SEAT No. :

PC-2369

[Total No. of Pages : 2

[6354] 486

B.E. (Computer Engineering)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410242)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 OR Q.2, Q.3 OR Q.4, Q.5 OR Q.6, Q.7 OR Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Explain Lasso Regression. Explain how Lasso Regression is used for feature selection. **[6]**

b) Define different regression models. **[6]**

c) Describe the bias-variance trade-off and its relationship to under fitting and overfitting. **[6]**

OR

Q2) a) Explain the advantages of RMSE over MSE as an evaluation metric. **[6]**

b) What do you mean by least square method? Explain least square method in the context of linear regression. **[6]**

c) Write a short note on Stochastic gradient descent algorithms. **[6]**

Q3) a) Explain kernel methods which are suitable for SVM. **[6]**

b) What are advantages and disadvantages of K-NN? **[6]**

c) What are different distance metrics are used in K-NN? **[5]**

P.T.O.

OR

Q4) a) What is Multi Class Classification? Explain the variants of Multi Class Classification. [5]

b) What are different techniques used for outlier handling? [6]

c) With suitable diagram, Explain Random forest Algorithm with example. [6]

Q5) a) Why K-medoid is used? Explain K-medoid algorithm. [5]

b) Why density based clustering is used? Explain any one. [6]

c) Cluster the following eight points (with (x, y) representing locations) into three clusters. [6]

P1(1, 3), P2(2, 2), P3(5, 8), P4(8, 5), P5(3, 9), P6(10, 7), P7(3, 3), P8(9, 4), P9(3, 7)

Use K-Means Algorithm to find the three cluster

OR

Q6) a) What is isolation factor model? [5]

b) Explain Hierarchical Clustering with an example. [6]

c) Micro-Average Precision and Recall, Micro-Average F-score, [6]

Q7) a) Explain Recurrent Neural Networks with an example. [6]

b) What are different activation function used in NN? [6]

c) What is multilayer perceptron? Describe with diagram. [6]

OR

Q8) a) Explain building blocks of RBF networks. [6]

b) What is personalized recommendation? What is content based recommendation? [6]

c) Explain the Convolution Neural Network (CNN) with suitable example. [6]

